Mathematical Analysis — Exercise Workbook

29 October 2026 · Răzvan

How to use this workbook

All exercises from the five official seminar sheets (analysis_seminar1–5.pdf, Class Materials → Seminars), numbered exactly as on the sheets.

Tick each box once you can solve the exercise without looking at the solution.

Mark	Meaning
(S)	solved at the seminar (it appears in my notes)
(S, partial)	started at the seminar, not finished
(S, hint)	only a hint was given at the seminar
(HW)	left as homework
—	not done at the seminar

C = convergent, D = divergent (the sheets' own convention). $\mathbb{N} = \{1, 2, 3, \ldots\}$, as used at the seminar.

Required solution format: the seminar sheets give no explicit format. The solutions document follows the seminar: state the method or test, compute step by step with $\Longrightarrow$ between steps, then write the conclusion on its own line.

Full solutions and the theory for each chapter are in *Mathematical Analysis — Solutions & Theory*.

Method finder

If the exercise has…	Try first	Chapter
a set, "find sup / inf / max / min"	draw it on the real line, find lb and ub	1
"neighborhood", "interior", "closure"	look for an interval $(x - \varepsilon, x + \varepsilon)$	1
$\sqrt{(n+1)} - \sqrt{n}$	multiply by the conjugate	2
a ratio of polynomials in n	factor out the highest power of n	2
a 1^∞ form, (1 + something small)^(something large)	the number e	2, 5
a sum of n terms divided by something	Stolz–Cesàro	2

If the exercise has…	Try first	Chapter
factorials n! or products 1·3·…·(2n−1)	ratio test	2, 3
(…)^n or (…)^(n^2)	root test	3
n to a constant power, ln n	comparison with a p-series	3
ln n together with n in the denominator	condensation test	3
$(-1)^n$	Leibniz test, then absolute convergence	4
a power series in (x − c)	ratio formula for R, then test the endpoints	4
a product of functions, n-th derivative	Leibniz rule	5

Chapter 1 — Real numbers (Seminar 1)

1.1 Bounds, sup, inf, max, min

Method: find lb(A) and ub(A) first, then read off the four values.

Ex. 1 Find the lower and upper bounds, then sup, inf, max, min.

- [ ] (a) **(S)**

$[-3, 2) \cup \{3\}$

- [ ] (b) **(S)**

$(-1, 1] \cup (2, \infty)$

- [ ] (c) **(S)**

$(-5, 5) \cap \mathbb{Z}$

- [ ] (d)

$\emptyset$

Ex. 2 Find sup, inf, max, min.

- [ ] (a) **(S)**

$\{x \in \mathbb{Q} \mid x^2 < 3\}$

- [ ] (b) **(S)**

$\{x^2 - 4x + 3 \mid x \in \mathbb{R}\}$

☐ (c) **(S)**

$$\left\{ \frac{n}{n+1} \;\middle|\; n \in \mathbb{N} \right\}$$

☐ (d)

$$\{2^{-k} + 3^{-m} \mid k, m \in \mathbb{N}\}$$

1.2 Proofs with sup and inf

Method: use the two defining properties of sup / inf — it is a bound, and it is the best bound.

☐ **Ex. 3 (S, partial)** S nonempty and bounded above. Show that −S is bounded below and

$$\inf(-S) = -\sup(S)$$

☐ **Ex. 4 (S: inf part and example; sup part HW)** For f, g : D → ℝ, D ≠ ∅, prove the two inequalities and give examples where they are strict:

$$\inf_{x\in D}\big(f(x)+g(x)\big) \geq \inf_{x\in D} f(x) + \inf_{x\in D} g(x), \qquad \sup_{x\in D}\big(f(x)+g(x)\big) \leq \sup_{x\in D} f(x) + \sup_{x\in D} g(x)$$

☐ **Ex. 5** a > 0, S nonempty and bounded above. Prove that

$$\sup_{x\in S}(ax+b) = a\sup(S) + b$$

1.3 Neighborhoods

Method: find ε > 0 with (x − ε, x + ε) ⊆ V.

☐ **Ex. 6 (S)** Which of the following sets are neighborhoods of 0?

$$[-1,1]\cup\{2\}; \qquad (-1,1)\cap\mathbb{Q}; \qquad \bigcap_{n=1}^{\infty}\left[-\frac{1}{n},\frac{1}{n}\right]$$

☐ **Ex. 7 (S)** Let x ∈ ℝ. Prove that

$$U, V \in \mathcal{V}(x) \implies U \cap V \in \mathcal{V}(x)$$

☐ **Ex. 8** Let a, b ∈ ℝ, a ≠ b. Prove that there exist neighborhoods

$$U \in \mathcal{V}(a),\ V \in \mathcal{V}(b) \ \text{ such that } \ U \cap V = \emptyset$$

1.4 Interior and closure

Method: int = points with a whole interval inside A; cl = points such that every interval around them meets A.

Ex. 9 Find the interior and the closure.

☐ (a) **(S)**

$(1, 2]$

☐ (b) **(S)**

$[-3, 2) \cup \{3\}$

☐ (c) **(S)**

$(-1, 1] \cup (2, \infty)$

☐ (d) **(S)**

$(-5, 5) \cap \mathbb{Z}$

☐ **Ex. 10** Let A = (0, 1) ∩ ℚ. Show rigorously that inf A = 0, sup A = 1, int A = ∅ and cl A = [0, 1].

Chapter 2 — Sequences (Seminar 2)

2.1 The ε-definition of the limit

- [ ] **Ex. 1 (S)** Prove using the ε-definition that

$$\lim_{n\to\infty} \frac{1}{\sqrt{n}} = 0$$

2.2 Bounded + monotone ⟹ convergent (Weierstrass)

Ex. 2 Study if the sequence (x_n) is bounded, monotone and convergent.

- [ ] (a) **(S, HW)**

$$x_n = \sqrt{n+1} - \sqrt{n}$$

- [ ] (b) **(S)**

$$x_n = \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \cdots + \frac{1}{n(n+1)}$$

- [ ] (c) **(S)**

$$x_n = \frac{2^n}{n!}$$

- [ ] **Ex. 4 (S, partial)** Prove that the sequence below is increasing and bounded, hence convergent. Its limit is denoted by e.

$$e_n = \left(1 + \frac{1}{n}\right)^n$$

2.3 Computing limits

Methods: conjugate, highest power of n, the number e, factor out the largest term, squeeze.

Ex. 3 Find the limit.

- [ ] (a) **(S)**

$$\sqrt{n}\left(\sqrt{n+1} - \sqrt{n}\right)$$

- [ ] (b)

$$\frac{2^n + (-1)^n}{3^n}$$

☐ (c) **(S)**

$\sqrt[n]{n}$

☐ (d) **(S)**

$$\frac{(an+1)^2}{4n^2-2n+1}, \quad a \in \mathbb{R}$$

☐ (e) **(S)**

$$\left(a_1^n + a_2^n + \cdots + a_k^n\right)^{\frac{1}{n}}, \quad a_i > 0$$

Ex. 5 Find the limit.

☐ (a) **(S)**

$$\left(\frac{2n+1}{2n-1}\right)^n$$

☐ (b) **(S)**

$$n\big(\ln(n+2) - \ln(n+1)\big)$$

2.4 Stolz–Cesàro and the ratio–root link

☐ **Ex. 6 (a) (S)** Let (x_n) be convergent. What can you say about the averages below? Can the averages converge, but the sequence not?

$$a_n = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

☐ **Ex. 6 (b) (S)** Compare 1 + 1/2 + ... + 1/n with n and with ln n, by taking the ratio.

☐ **Ex. 7 (S, partial — HW)** Prove that

$$\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = \ell \implies \lim_{n\to\infty} \sqrt[n]{x_n} = \ell$$

Ex. 8 Find the limit.

☐ (a) **(S, hint)**

$$\frac{n}{\sqrt[n]{n!}}$$

☐ (b) **(S, hint)**

$$\frac{1^p + 2^p + 3^p + \cdots + n^p}{n^{p+1}}, \quad p \in \mathbb{N}$$

Chapter 3 — Series with positive terms (Seminar 3)

3.1 Sums of series

Methods: geometric series, the series of e, telescoping.

Ex. 1 Find the sum.

- [ ] (a) **(S)** — geometric series

$$\sum_{n\geq 1} \frac{2}{3^n}$$

- [ ] (b) — split, then use the series of e

$$\sum_{n\geq 1} \frac{2n+1}{n!}$$

- [ ] (c) — partial fractions, telescoping

$$\sum_{n\geq 1} \frac{1}{4n^2-1}$$

- [ ] (d) **(S)** — telescoping

$$\sum_{n\geq 1} \frac{1}{n(n+1)(n+2)}$$

- [ ] (e) — differentiated geometric series

$$\sum_{n\geq 1} \frac{n}{2^n}$$

- [ ] (f) — comparison; the sum is $+\infty$

$$\sum_{n\geq 1} \frac{1}{\sqrt[3]{n}}$$

3.2 Comparison tests

Ex. 2 Study the convergence.

- [ ] (a) **(S)** — compare with the harmonic series

$$\sum_{n\geq 2} \frac{1}{\ln n}$$

- [ ] (b) **(S, partial)** — compare with a p-series, p = 3/2

$$\sum_{n\geq 1} \frac{1}{n\sqrt{n+1}}$$

- [ ] (c) **(S)** — use ln(1 + x) < x

$$\sum_{n\geq 1} \frac{\ln\left(1+\frac{1}{n}\right)}{n}$$

3.3 Ratio and root tests

- [ ] **Ex. 2 (d)** — ratio test

$$\sum_{n\geq 1} \frac{n!}{(n+1)^n}$$

- [ ] **Ex. 2 (e) (S)** — root test

$$\sum_{n\geq 1} \left(\frac{n}{n+1}\right)^{n^2}$$

3.4 Condensation test

- [ ] **Ex. 2 (f) (S)**

$$\sum_{n\geq 2} \frac{1}{n\ln n}$$

3.5 Tests with a parameter; when the ratio test gives 1

Ex. 3 Study the convergence.

- [ ] (a) **(S)** — ratio gives 1, then Raabe–Duhamel

$$\sum_{n\geq 1} \frac{1\cdot 3\cdot \ldots \cdot (2n-1)}{2\cdot 4\cdot \ldots \cdot 2n}$$

- [ ] (b) **(S)** — rewrite as a p-series

$$\sum_{n\geq 1} a^{\ln n}, \quad a>0$$

- [ ] (c) **(S)** — ratio test, the case a = e separately

$$\sum_{n\geq 1} \frac{a^n\, n!}{n^n}, \quad a>0$$

Chapter 4 — Alternating series and power series (Seminar 4)

4.1 Convergence and absolute convergence

Method: Leibniz test for convergence; comparison on $|x_n|$ for absolute convergence.

Ex. 1 Study the convergence and the absolute convergence.

- [ ] (a)

$$\sum_{n\geq 1} \frac{(-1)^{n+1}}{\sqrt{n(n+1)}}$$

- [ ] (b)

$$\sum_{n\geq 1} (-1)^n \sin\frac{1}{n}$$

- [ ] (c)

$$\sum_{n\geq 1} \frac{\sin n}{n^2}$$

4.2 Sums from the geometric series

Method: differentiate or integrate the geometric series term by term.

- [ ] **Ex. 2 (S)** Prove by differentiating the geometric series that, for $|x| < 1$,

$$\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2}, \qquad \sum_{n=2}^{\infty} n(n-1)x^n = \frac{2x^2}{(1-x)^3}$$

- [ ] **Ex. 3 (S)** Prove by integrating the geometric series that, for $|x| < 1$,

$$\sum_{n=1}^{\infty} \frac{x^n}{n} = -\ln(1-x), \qquad \sum_{n=1}^{\infty} (-1)^{n+1}\frac{x^n}{n} = \ln(1+x)$$

- [ ] **Ex. 4 (S)** Prove that, for $x \in [-1, 1]$,

$$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots = \arctan x$$

4.3 Radius of convergence and convergence set

Method: compute R with the ratio (or root) formula, then test the two endpoints separately.

Ex. 5 Find the radius of convergence and the convergence set.

☐ (a) **(S)**

$$\sum_{n\geq 0} \frac{x^n}{2^n}$$

☐ (b) **(S, partial)**

$$\sum_{n\geq 0} 2^n (x-1)^n$$

☐ (c) **(S)**

$$\sum_{n\geq 1} \frac{(x-2)^n}{(n+1)3^n}$$

☐ (d) **(S)**

$$\sum_{n\geq 0} n!\, x^n$$

☐ (e) **(S)**

$$\sum_{n\geq 1} \frac{(x-1)^n}{n^p}, \quad p>0$$

☐ (f) **(S, partial)**

$$\sum_{n\geq 1} \frac{(-1)^n x^n}{\sqrt{n}}$$

Chapter 5 — Limits of functions, continuity, derivatives (Seminar 5)

5.1 Accumulation points

☐ **Ex. 1** Find the accumulation points of each set:

$$[0,1)\cup\{2\},\qquad \mathbb{Z},\qquad \mathbb{Q},\qquad \{0.1,\ 0.11,\ 0.111,\ \dots\}$$

5.2 Continuity

Methods: the sequential criterion, the intermediate value theorem.

☐ **Ex. 2** Find a function f : ℝ → ℝ that is discontinuous everywhere, with |f| continuous everywhere.

☐ **Ex. 3** Prove that a continuous function f : [a, b] → [a, b] has at least one fixed point x* = f(x*).

☐ **Ex. 4** Study the continuity and the differentiability of f and f′, where

$$f(x)=\begin{cases} x^2\sin\frac{1}{x}, & x\neq 0\\ 0, & x=0\end{cases}$$

5.3 Fundamental limit and derivatives from the definition

☐ **Ex. 5 (S, partial)** Prove from scratch the limit below; then prove that (sin x)′ = cos x and (cos x)′ = −sin x.

$$\lim_{x\to 0}\frac{\sin x}{x}=1$$

5.4 Limits of functions

Methods: squeeze, the number e, the identity u^v = e^(v ln u), L'Hôpital, Taylor.

Ex. 6 Compute the limits.

☐ (a)

$$\lim_{x\to\infty}\frac{\lfloor x\rfloor}{x}$$

☐ (b)

$$\lim_{x\to\infty} x\big(\ln(x+2)-\ln(x+1)\big)$$

☐ (c) **(S)**

$\lim_{x\to 0} (\cos x)^{\frac{1}{x^2}}$

☐ (d)

$\lim_{x\to 0,\ x>0} x^x$

☐ (e)

$\lim_{x\to 0,\ x>0} (\sin x)^x$

☐ (f)

$\lim_{x\to\infty} x\left(\left(1+\frac{1}{x}\right)^x - e\right)$

5.5 The n-th derivative

Method: find the pattern from f′, f″, f‴ and prove it by induction; for products use the Leibniz rule.

Ex. 7 Find the n-th derivative.

☐ (a) **(S, partial)**

$f : (-1, \infty) \to \mathbb{R}, \quad f(x) = \ln(1+x)$

☐ (b)

$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = \sin x$

☐ (c)

$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = x^2 \sin x$

☐ (d)

$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = e^{2x} x^3$